

ference isn't limited to panel type alone. Even the grade and manufacturer of the panels matter a lot. Both IPS and VA panel technology has improved over the last several years, working on perfecting its strengths and eliminating its weaknesses. As a result, the actual difference can only be judged by taking a demo of the TV and comparing various models to arrive at the better panel. The ultra affordable TVs often tend to use lower quality panels to save cost. Having said that, there are exceptions in the market who claim to use A+ grade IPS panels. However, most low cost TVs with good quality IPS panels cut corners over LED backlight implementations, they usually offer Direct LED based backlights with very few LED clusters at the back, impacting contrast performance and ending up with unevenly lit panels as well.

Number of HDMI ports:

Look for a TV with a bare minimum of 2 HDMI ports. Though, we recommend you to opt for a slightly more future proof solution that offers 3-4 HDMI ports, ensuring you can connect more devices to your TV.

Refresh rate:

As explained earlier at the start of this guide, refresh rate is particularly a crucial factor for low-cost 4K TVs, with some of them delivering a rather low 30hz refresh rate at 4K resolutions. Most 1080p TVs offer 50/60 hz refresh rates ensuring smooth playback of video that is shot at 60 frames per second. A lower refresh could cause judder and choppiness during video playback due to the inability of the panel to refresh quickly enough as per the video being fed to the panel. While most movies were earlier shot at 24/30 frames per second, sports channels and even some of the latest movies are shot at 48FPS, thus further validating the need for a TV that has a panel with minimum 60hz refresh rate being offered at all supported resolutions. Also, one should not confuse the often quoted and heavily marketed terms like "Ultra Motion" or "Motion Clarity Index" with refresh rates, these aren't native or real refresh rates but forced refresh rates derived by using frame interpolation methods. These implementations hardly make any difference and in fact may sometimes even degrade the image quality.

Panel glass finish:

LED TVs come with two kinds glass finishes, glossy or matte (semi-glossy). The glossy finish offers slightly more contrast at the cost of causing ugly reflections. We recommend you opt for matte panel based TVs unless your viewing setup is well controlled in terms of lighting.

HDMI Return Channel/ HDMI ARC Support:

This is an important feature if you plan to connect your home theater/receiver to your TV, with the HDMI Audio Return Channel port, you need fewer cables and can manage volume levels of all multiple input devices via a single remote.

Contrast Ratio:

While contrast ratio is a very important specification for a TV, the insanely high quoted figures along with confusing and arbitrary labels such as 'dynamic' or 'mega' contrast are fake and should never be taken on face value. We suggest you



ignore all contrast ratio claims made by manufacturers unless you come across a number that reads more like 1000:1 or 2000:1. Very few TVs can manage an actual contrast ratio that goes beyond 1000:1. Also, the measuring criteria and environment in terms of brightness levels and calibration of a TV play a very important role in the process of contrast ratio measurement, hence it is recommended you compare contrast ratios rather than taking any absolute figure.

USB connectivity and playback support:

Most TVs offer USB 2.0 ports, even the basic inexpensive models. USB ports

allow you to play media files directly via USB based storage devices. 4K TVs offer USB 3.0 ports for higher bandwidth video transfer.

Built-In WiFi, Miracast/ WiFi direct:

Given the fact that smartphones are at the core of most peoples tech ecosystem today, support for beaming content directly to the TV and screen mirroring ability can be more than handy. A lot of TVs that aren't exactly Smart TVs in the true sense do offer basic WiFi and Miracast features, if you aren't particularly interested in a Smart TV, we suggest you look for TVs that offer these basic wireless connectivity.

Speakers position, rating:

A TV with 20W RMS sound output is the bare minimum that you should settle for. Apart from the output, the position and direction of sound matters a lot. Front facing speakers will offer

better sound quality in terms of fidelity and clarity over rear or down facing speakers. Unfortunately, the craze for slimmer, narrower bezels and minimal designs has led to manufacturers opting for back firing or down firing speakers where the drivers are placed at the bottom (facing the ground) or at the back (facing the wall) to maintain the minimalist look. As a result, sound quality does get heavily compromised. Also, be wary of fancy 'dynamic' or 'virtual' surround sound claims by TVs, such gimmicks don't deserve a premium and should be set turned off to ensure you get the most 'clean' sound that the TV is capable of. **[E]**